
From Structural to Temporal Fidelity in Dynamic Graph Generation

Abstract

Language model agents are increasingly used to simulate dynamic text attributed graphs, but it is unclear whether more efficient systems preserve the temporal dynamics of real interaction streams. Existing evaluations mainly measure *state fidelity* through structural, textual, and marginal timing statistics, while applications such as forecasting and intervention analysis also require *process fidelity*. We show that statistics based only on marginal behavior cannot distinguish a process that preserves event ordering from one that destroys temporal dependence. We therefore introduce R2O, a likelihood-free framework that evaluates temporal fidelity along three dimensions. *Rate* measures the pace of activity, *Rhythm* measures the distribution of inter event times, and *Order* measures dependence between consecutive intervals. Each discrepancy is calibrated against variation between matched windows of real data. Across two agent architectures, three backbone families, and four datasets, R2O reveals a consistent separation between structural and temporal fidelity. Configurations that improve structural realism do not consistently move closer to the real temporal process, and no pre-registered configuration reproduces the temporal dependence observed in the real streams. Changing the language model backbone changes which temporal properties are better matched, but does not yield consistent agreement across Rate, Rhythm, and Order. These results show that structural fidelity alone is insufficient for evaluating efficient dynamic graph generation.

1 Introduction

Language model agents are increasingly used to simulate evolving social and interaction systems represented as dynamic text attributed graphs [Zhang et al., 2024, Ji et al., 2025, Peng et al., 2025, Ji et al., 2026]. Such systems combine language model decisions with interaction history, memory, and tool use to generate both graph structure and text over time. Simulation over large populations and long horizons makes inference cost a central design constraint, which motivates the use of smaller language model backbones [Belcak et al., 2025, Yang et al., 2025]. In practice, reduced backbone capacity can also be paired with richer agent scaffolding. This creates an important question for small model agent systems. Whether a cheaper backbone, additional scaffolding, or their combination preserves the process that the agents are intended to simulate cannot be determined from efficiency or output realism alone.

This issue is different from asking whether a small model can reproduce the same individual decisions or final graph statistics as a larger model. In an agent simulation, backbone and scaffold choices are repeatedly applied as the system evolves, so apparently local differences can alter when interactions occur and how subsequent activity depends on prior events. A configuration can therefore produce plausible agents and realistic graph structure while generating a different temporal process. For efficient agent systems, the relevant tradeoff is consequently not only quality against computational cost, but fidelity of the generated process against computational cost. Evaluating that tradeoff requires

measurements that remain meaningful when backbone and scaffold choices change the dynamics without substantially changing the resulting states.

Existing graph generation evaluation does not directly provide such a test. Static graph generators are primarily compared through structural statistics such as degree, clustering, and motifs [You et al., 2018, Liao et al., 2019, O’Bray et al., 2022]. DyTAG systems extend this evaluation to generated text, activity trajectories, and marginal timestamp statistics [Ji et al., 2025, Peng et al., 2025, Ji et al., 2026]. Related language model social simulations have begun to model temporal behavior more explicitly. These approaches demonstrate the importance of temporal behavior, but the literature summarized in Appendix B does not provide an evaluation framework for LM agent DyTAG streams that jointly tests order sensitive event dependence and interprets the resulting discrepancy relative to variation between comparable real streams.

We refer to agreement on generated structure, text, and marginal activity statistics as *state fidelity*. For many uses of simulation, matching the state is not sufficient. Forecasting, intervention analysis, and synthetic event generation depend on how future events vary with preceding history. We call agreement in this conditional evolution *process fidelity*. For a graph sequence $G_{1:T}$, the distinction is between properties of the marginal distribution $P(G_t)$ and properties of the conditional distribution $P(G_t | G_{<t})$. This distinction is particularly relevant when comparing small model backbones and agent scaffolds because two configurations can agree on the states they produce while differing in how those states evolve.

Marginal agreement does not identify temporal dependence. Permuting an actor’s inter event times preserves its event count, inter event time distribution, and statistics computed from the time aggregated graph while changing the order in which activity unfolds. We formalize this observation and show that statistics depending only on the aggregated graph or unordered inter event times are invariant to this transformation. Process fidelity therefore requires at least one statistic that responds to event order.

We introduce R2O, a likelihood free framework that evaluates temporal process fidelity along three separate axes. *Rate* measures activity pace, *Rhythm* measures the inter event time distribution, and *Order* measures dependence between consecutive intervals. Each discrepancy is interpreted relative to variation between matched windows of the real stream. We establish order sensitivity, separation among the three axes, and finite sample calibration, and validate these properties with shuffled and memoryless controls. Proofs and synthetic validation are provided in Appendix C.

Our contributions are (1) a formal distinction between state and process fidelity for LM agent DyTAG simulation, together with an invariance result showing why structural and marginal timing statistics cannot detect loss of event order, (2) R2O, which separately evaluates activity rate, inter event time distributions, and temporal dependence relative to variation within the real data, and (3) an evaluation across agent architectures, backbone families, and datasets that tests whether structural improvements obtained through backbone and scaffold choices also preserve temporal dynamics. Empirically, structural improvements do not reliably reduce temporal discrepancies, marginal timing agreement need not imply agreement in event dependence, and changing the language model backbone does not consistently resolve the gap across Rate, Rhythm, and Order. These results show why process fidelity is a necessary consideration when evaluating efficient agent simulation.

2 Requirements for measuring process fidelity

Temporal process fidelity requires more than agreement in aggregate properties. An evaluation must respond to event order, distinguish sources of temporal error, and provide a scale on which discrepancies can be interpreted.

2.1 Event streams and classes of statistics

An event stream is $E = ((u_i, v_i, x_i, t_i))_{i \leq n}$, where u_i and v_i denote the interacting actors, x_i is the associated text, and t_i is a non decreasing timestamp. For actor u , let $\tau_j^u = t_{j+1}^u - t_j^u$ denote its j th

inter event time (IET), T^u the multiset of its IETs, $T = \biguplus_u T^u$ the pooled IET multiset, and $\bar{G}(E)$ the time aggregated graph.

Definition 1 (State, marginal timing, and dependence statistics). *A statistic is a state statistic if it is computed from generated states without using the order of an actor’s IETs. It is a marginal timing statistic if it depends on timing only through the multisets $\{T^u\}_u$, and a dependence statistic if changing the order of IETs within an actor can change its value.*

Structural statistics such as degree MMD, clustering, and motifs are widely used to compare generated and real graphs [You et al., 2018, Liao et al., 2019, O’Bray et al., 2022]. DyTAG generators also compare text similarity, activity trajectories, and count based temporal summaries [Ji et al., 2025, Peng et al., 2025, Ji et al., 2026]. These are state statistics because they do not use IET order. Pooled IET KS, W_1 on log IETs, the burstiness gap $|\Delta B|$, and node rhythm divergence are marginal timing statistics [Goh and Barabási, 2008, Karsai et al., 2012]. Memory M and global lag 1 IET autocorrelation are dependence statistics because they use consecutive intervals. We use *state fidelity* for agreement on state and marginal timing properties and *process fidelity* for agreement in temporal dependence. A broader comparison is in Appendix B.

2.2 Requirement 1: sensitivity to event order

Agreement in marginal timing does not determine temporal dependence. Two processes can have the same IET distributions and time aggregated graph statistics while differing in how successive events depend on their history.

Proposition 1 (Non identification of temporal dependence). *Let F be any IET distribution with finite variance. Let P_0 generate each actor’s IETs independently from F , and let P_ρ generate them from a stationary Gaussian copula $AR(1)$ process with the same marginal F and parameter $\rho \in (0, 1)$. If partner and text choices are identical and independent of event timing, then P_0 and P_ρ have identical actor level IET marginals, identical burstiness, and identical distributions for every state statistic, while $M(P_\rho) > 0 = M(P_0)$.*

The same limitation can be exposed directly on an observed stream. Let $\varsigma(E)$ independently permute the IET sequence of each actor.

Proposition 2 (Shuffle invariance). *Statistics that depend only on $\bar{G}(E)$ or on the unordered IET multisets $\{T^u\}_u$ are unchanged by ς . For an actor with $k_u \geq 3$ IETs,*

$$\mathbb{E}[M^u(\varsigma(E)) \mid E] = O(1/k_u).$$

Thus, a generator can reproduce each actor’s IET marginal while carrying no temporal memory and remain indistinguishable under order invariant statistics.

Process evaluation must therefore include at least one statistic that changes when event order changes. The construction and proofs of both propositions are provided in Appendix C.

2.3 Requirement 2: separation of failure modes

Temporal error can arise in different ways. A simulator may generate events at the wrong overall rate, reproduce the wrong distribution of inter event times, or fail to reproduce dependence between consecutive intervals. These errors cannot be diagnosed reliably after they are collapsed into a scalar.

Proposition 3 (Separation under rate changes). *Under a global rescaling $\tau \mapsto a\tau$, burstiness B , memory M , and lag 1 autocorrelation are invariant. Pooled IET KS and node rhythm KS are not invariant, and*

$$|W_1(\log a\tau, \log \tau') - W_1(\log \tau, \log \tau')| \leq |\log a|.$$

Equality holds when the two distributions have the same shape. Hence $\log(T_g/T_r)$ isolates rate, B and $W_1(\log \tau)$ characterize marginal timing up to the stated rate term, and M and lag 1 autocorrelation characterize dependence.

Rate, marginal timing, and temporal dependence should therefore be reported separately. These three failure modes define the *Rate*, *Rhythm*, and *Order* axes of R2O. The proof is given in Appendix C.

2.4 Requirement 3: calibration against real data variability

Raw discrepancy values do not have a common interpretation across datasets or evaluation windows. The same KS distance, Wasserstein distance, or change in memory may reflect ordinary variation in one dataset but a mismatch in another. Standard point process diagnostics can obtain a reference distribution from a specified conditional intensity [Ogata, 1988, Brown et al., 2002], but the LM agent DyTAG generators considered here do not expose a tractable conditional event likelihood [Ji et al., 2025, Peng et al., 2025, Ji et al., 2026]. We therefore interpret each simulator to real discrepancy relative to the same discrepancy measured between matched windows of the real stream.

For a simulator to real discrepancy D^{SR} , let $D_{1:N}^{RR}$ denote the same discrepancy computed between N matched pairs of real windows and define

$$q_{RR} = \frac{1 + \#\{i : D_i^{RR} \geq D^{SR}\}}{N + 1}.$$

Proposition 4 (Finite sample calibration). *If D^{SR} is exchangeable with $D_{1:N}^{RR}$, then $\Pr(q_{RR} \leq \alpha) \leq \alpha$ for every $\alpha \in [0, 1]$. Moreover, $q_{RR} = 1/(N + 1)$ exactly when D^{SR} exceeds every real to real discrepancy.*

The calibration places each simulator discrepancy on the scale of variability already present in the observed process. R2O applies it separately to each axis rather than combining heterogeneous discrepancies into a single score. The proof and matching details are given in Appendices C and D.

These requirements define how R2O should distinguish different forms of temporal mismatch. Structural improvements need not reduce temporal discrepancies, agreement on Rhythm need not imply agreement on Order, and the three axes may favor different configurations. A discrepancy should also remain outside the matched real reference after accounting for plausible measurement artifacts. Section 4 evaluates these patterns empirically.

3 The R2O framework

The requirements above determine the structure of R2O. Given a generated stream E_g and a corresponding real stream E_r , the framework computes temporal discrepancies along Rate, Rhythm, and Order and calibrates each one against comparable real data segments.

3.1 Evaluation procedure

R2O can compare complete streams or matched continuations following a common observed history. It does not prescribe how E_g is produced and can therefore score both conditioned and closed loop generation. Generation and evaluation remain separate, allowing the same temporal criteria to be applied across simulator designs. For each generated and real pair, the statistics below are assigned to an axis and calibrated against the matched real reference.

3.2 Rate, Rhythm, and Order

Rate Rate measures agreement in the pace of activity. For generated and real durations T_g and T_r ,

$$D_{\text{Rate}} = \left| \log \frac{T_g}{T_r} \right|.$$

The discrepancy is zero when the two time scales agree and changes under uniform rescaling while the Order statistics remain invariant.

Rhythm Rhythm measures agreement in the marginal distribution of IETs without using their sequential order. At the population level, R2O uses pooled IET KS, Wasserstein distance on log IETs, and the burstiness difference $|\Delta B|$ [Goh and Barabási, 2008, Karsai et al., 2012]. At the actor level, node rhythm divergence is

$$D_{\text{node}} = \text{median}_u \text{KS}(T_g^u, T_r^u).$$

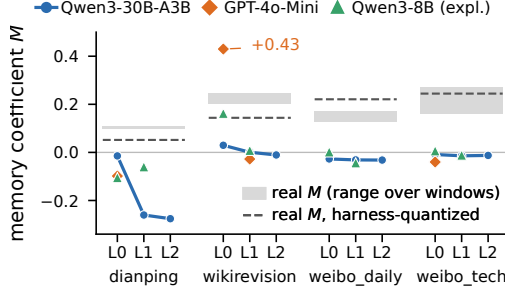


Figure 1: Temporal dependence across configurations. Further results in Appendix E.

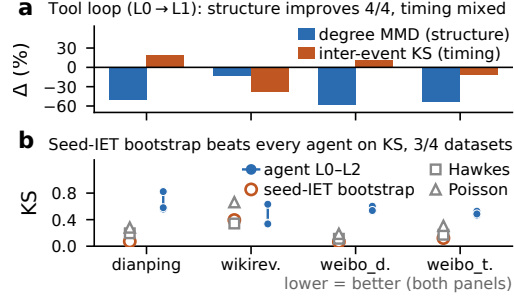


Figure 2: Structural and temporal responses. Complete values reported in Appendix E.

171 The pooled statistics characterize population timing, while node rhythm preserves actor identity and
 172 can expose discrepancies hidden by pooling.

173 **Order** Order measures agreement in dependence between consecutive intervals. R2O uses $D_M =$
 174 $|M_g - M_r|$ and $D_\rho = |\rho_{1,g} - \rho_{1,r}|$. Unlike Rhythm, these statistics change when same intervals are
 175 reordered. They distinguish streams that share an IET distribution but differ in temporal dependence,
 176 which is the failure mode established by Propositions 1 and 2. R2O does not combine Rate, Rhythm,
 177 and Order into a scalar because agreement on one axis need not imply agreement on others.

178 3.3 Calibration with real data

179 Each discrepancy is interpreted relative to matched real segments. The same R2O statistic is computed
 180 across observed window pairs to form a real reference distribution, and the simulator is located within
 181 its upper tail using q_{RR} . A small value indicates a discrepancy larger than typical variation in the real
 182 process. Matching and controls for timestamp ties, scheduler quantization, actor activity, and window
 183 selection are described in Appendix D.

184 **Construction of the real reference** R2O builds the reference from pairs of 10,000 edge real windows
 185 matched on duration, actor count, edges per actor, and median positive IET, using up to 60 pairs
 186 per dataset with reuse of any window capped at three. We report q_{RR} together with a standardized
 187 discrepancy Z_D in units of real variation. The calibration compares each simulator against comparable
 188 real activity regimes rather than arbitrary windows. Full details are in Appendix D.

189 **Synthetic validation of axis separation** Synthetic controls verify the intended separation before
 190 model evaluation. Shuffling a dependent stream preserves its IET distribution and burstiness while
 191 reducing memory from 0.164 to -0.007 ; an independent renewal process with the same marginal
 192 behaves similarly. Rescaling time by a factor of 20 leaves memory at 0.164 while moving IET KS to
 193 0.686 and log Wasserstein distance to 2.996. The complete validation is in Appendix C.

194 4 Experiments and findings

195 We evaluate R2O on dianping, wikirevision, weibo_daily, and weibo_tech, the four GDGB
 196 datasets with suitable temporal information [Peng et al., 2025]. GRAPHIA is excluded from event
 197 timing experiments because it generates in fixed daily intervals without continuous event timestamps
 198 (Appendix F). The primary experiments compare L0, which reproduces GAG General [Ji et al.,
 199 2025], L1, which adds tool use, and L2, which additionally includes memory. Qwen3 30B A3B is
 200 the primary backbone, GPT 4o Mini provides matched backbone comparisons, and Qwen3 8B is
 201 exploratory. The primary analysis contains 16 pre registered configurations.

202 Each run starts from a real prefix and is scored against the corresponding real continuation. We also
 203 include Poisson, empirical IET bootstrap, and Hawkes timing baselines. For Qwen3 8B, L0 uses
 204 1.9 to 2.8 language model calls per event and L1 uses 2.5 to 4.7, with 1.6 to 2.4 times more prompt
 205 tokens. Full dataset, generation, serving, baseline, and efficiency details are provided in Appendix A.

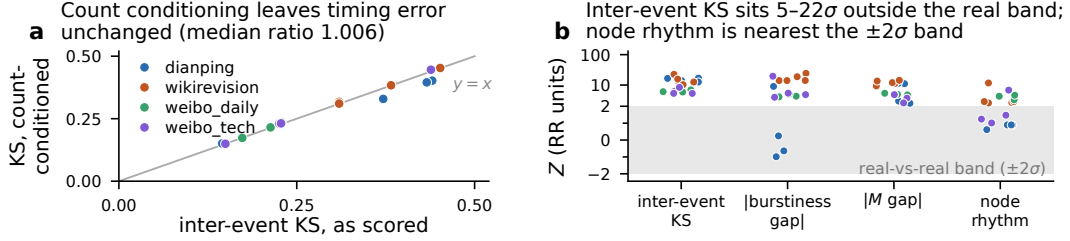


Figure 3: **R2O under measurement controls.** Count matching leaves marginal timing largely unchanged, while standardized temporal discrepancies remain outside the matched real reference.

R2O separates structural improvement from process fidelity From L0 to L1, degree MMD improves on all four datasets by 43.8% on average, while inter event KS changes by only 4.8% and not consistently (Figure 2). Even the best structural configuration on each dataset remains outside the corresponding real reference on temporal measures. On the representative wikirevision window (GPT 4o Mini; Appendix D), the log duration ratio also worsens from 3.83 to 4.44, or approximately 46 to 85 times the real duration. R2O therefore shows that structural improvement does not imply improved temporal process fidelity.

The Order axis detects discrepancies that Rhythm cannot identify Real memory coefficients range from 0.10 to 0.27, but none of the 16 pre registered configurations reproduces this dependence. Across the 12 Qwen3 30B A3B configurations, memory ranges from -0.28 to 0.03 , and every configuration lies outside the real reference range on the memory discrepancy (Figure 1). Rhythm fails less uniformly: actor level rhythm divergence lies inside the real reference on 7 of the 16 configurations, so a simulator can approach actors’ IET distributions without reproducing dependence between successive intervals. This is the distinction for which Order is needed.

The R2O axes reveal different effects of the language model backbone Matched Qwen3 30B A3B and GPT 4o Mini comparisons improve some temporal measures while worsening others, and exploratory Qwen3 8B runs show the same pattern. The only 8B configuration approaching real memory also has the largest marginal timing discrepancy in that set, and matched horizon controls place memory near the shuffled value in all eight exploratory arms. Backbone changes therefore alter which temporal properties are matched without consistent improvement on Rate, Rhythm, and Order.

The timing baselines show why Rhythm alone is insufficient The empirical IET bootstrap samples intervals independently from the observed seed distribution, retaining marginal timing but not temporal dependence. It outperforms every agent configuration on inter event KS on three of four datasets (Figure 2), with values from 0.065 to 0.122 compared with 0.468 to 0.559 for the best agent. Because the bootstrap generates neither graph structure nor text, its role is diagnostic. It provides an empirical instance of Proposition 1: its global lag 1 autocorrelation is -0.002 to $+0.002$ (20 seeds) against real 0.22 to 0.36, so strong Rhythm coexists with complete failure on Order. Complete configuration level, backbone, and timing baseline results are in Appendix E.

R2O findings remain stable under measurement controls Matching actor level event counts has little effect on marginal timing, and the standardized R2O scores remain outside variation across matched real windows (Figure 3). The Order discrepancy also persists after controls for timestamp ties, scheduler quantization, and evaluation window selection. The separation detected by R2O is therefore not explained by the scoring procedure or timestamp representation. Complete controls, synthetic validation, and formal analysis are in Appendices D and C.

Limitations The evaluation covers one DyTAG system lineage and shared generation pipeline, with teacher forced generation and four datasets. The Qwen3 8B analysis is exploratory, available configurations are single archived runs rather than replicated generation seeds, and some terminate before the nominal horizon. API served backbones also carry provider side serving opacity. Real to real calibration measures variation across observed windows but does not estimate generation seed uncertainty, and q_{RR} should be interpreted as an empirical tail position rather than a hypothesis test value. Additional limitations and serving details are in Appendix I.

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A Experimental details and reproducibility

A.1 Datasets and configurations

The experiments use dianping, wikirevision, weibo_daily, and weibo_tech from GDGB. These are the four datasets used for the temporal analysis in the main paper. All R2O statistics are computed from event timestamps after the same preprocessing used by the generation pipeline. Dataset access and preprocessing follow the GDGB release.

The primary analysis contains 16 preregistered configurations. Twelve use Qwen3-30B-A3B and cover the four datasets under L0, L1, and L2. L0 reproduces GAG General, L1 adds tool use, and L2 additionally includes memory. Four archived GPT-4o-Mini configurations provide matched backbone comparisons on dianping L0, wikirevision L0 and L1, and weibo_tech L0. A separate exploratory analysis contains eight Qwen3-8B configurations covering L0 and L1 on all four datasets. Exploratory results are excluded from statements that refer to the 16 preregistered configurations.

A.2 Generation protocol

Each run conditions on a real prefix containing 1,000 edges and generates a continuation of approximately 9,000 events in rounds of 50 events. The main evaluation uses the corresponding real continuation from the inherited 10,000-edge evaluation window. Reference statistics are computed from the matched real slots of the same window.

Several archived runs terminate before the nominal horizon. The archived Qwen3-30B-A3B runs for weibo_tech L1, weibo_tech L2, and weibo_daily L2 are shorter than the nominal run. The weibo_daily L2 structural evaluation is excluded because the archived structural output is contaminated, while its timing statistics use the archived run produced before contamination. The GPT-4o-Mini run for weibo_tech L0 contains 200 generated events. Its timing statistics use the same matched-slot procedure at the available horizon, while structural statistics are unavailable because the structural evaluation lane was not run for such a short stream. Configuration-level numbers of scored events are reported in Table 9.

The exploratory Qwen3-8B L1 runs were stopped at round boundaries because of generation time. Their lengths range from 4,900 to 5,696 events. For comparisons between exploratory L0 and L1, we therefore also report a matched horizon analysis in which the corresponding L0 stream is truncated to the L1 event count and rescored using the same procedure.

A.3 Backbones, serving, and efficiency

Qwen3-30B-A3B is served locally with vLLM using FP8 and a fixed decoding configuration. GPT-4o-Mini is accessed through a version-pinned commercial API. The provider does not expose its serving stack or a determinism guarantee, so API runs are not used to estimate variance for the local model. The exploratory Qwen3-8B runs use a commercial API with provider reasoning disabled. Temperature is 0.3 and the maximum generated token count per call is 1,024.

For the exploratory Qwen3-8B grid, API accounting gives 1.9 to 2.8 LM calls per event for L0 and 2.5 to 4.7 calls per event for L1. L1 uses 1.6 to 2.4 $\times$ more prompt tokens per event than L0. The resulting cost is \$4.48 to \$7.77 per cell and \$46.78 for the complete exploratory grid. Per-call token usage and call counts are retained in the supplementary material.

A.4 Timing baselines

Three timing baselines use only the 1,000-edge seed prefix. The Poisson baseline estimates a constant event rate from the seed. The empirical IET bootstrap samples seed inter-event times independently with replacement. The Hawkes baseline fits an exponential Hawkes process to the seed by maximum

461 likelihood. These baselines generate event times only and do not generate graph structure or text.
462 Reported baseline means and standard deviations use 20 simulation seeds.

463 **A.5 Registration and reproducibility**

464 The primary configuration set and the main R2O analyses were preregistered. The Qwen3-8B arm
465 was added after registration and is labeled exploratory throughout the paper.

466 All analysis code, per configuration scores, matched real-window construction, the scheduler and
467 count-conditioning controls, figure and table generation scripts, and a verification gate that recomputes
468 every statistic reported in the paper from the committed score files are provided in the anonymized
469 supplementary material. The repository data are read only under the analysis scripts, and every figure
470 and table is regenerable from the committed score files. Dataset access follows the GDGB release.

471 The preregistration includes hypotheses, falsifiers, exclusions, a power plan, and stop rules, together
472 with the corresponding deviation log. The eight exploratory Qwen3-8B cells were generated with the
473 same harness, with provider reasoning disabled, temperature 0.3, and `max_tokens=1024`. Per-call
474 token usage was logged, and the per event accounting above is recomputed from those logs by the
475 verification gate.

476 **B Evaluation landscape**

477 Table 1 summarizes how prior graph generation, temporal graph generation, language-model-based
478 graph generation, and language-model-based social simulation studies evaluate generated data. The
479 table is included to place the R2O requirements in the broader evaluation landscape. It distinguishes
480 structural, textual, marginal-temporal, temporal-dependence, likelihood-based, and closed loop
481 evaluation. The main paper uses this comparison for positioning and does not rely on it for the formal
482 results.

483 **C R2O definitions, proofs, and synthetic validation**

484 **C.1 Metric definitions**

485 For an actor with IET sequence $\tau_1, \dots, \tau_k$, burstiness is

$$B = \frac{\sigma_\tau - \mu_\tau}{\sigma_\tau + \mu_\tau},$$

486 where μ_τ and σ_τ are the mean and standard deviation of the IETs. This definition follows the
487 standard burstiness formulation [Goh and Barabási, 2008]. The memory coefficient is the lag 1
488 Pearson correlation between consecutive IETs, computed within actors and pooled using the available
489 adjacent pairs. The use of consecutive intervals follows prior work on temporal memory [Goh and
490 Barabási, 2008, Karsai et al., 2012]. Global lag 1 autocorrelation applies the same dependence
491 calculation to the globally ordered IET sequence.

492 The Rhythm axis uses the KS distance between pooled generated and real IET distributions, the
493 Wasserstein distance between log IET distributions, the absolute burstiness difference, and node-
494 rhythm divergence. Node-rhythm divergence is the median actor specific KS distance between the
495 generated and real IET distributions for the same actor. Rate uses the absolute log-duration ratio.
496 Order uses the absolute memory difference and the absolute difference in global lag 1 autocorrelation.

497 These statistics keep the three R2O axes distinct. Rate responds directly to a change in time scale.
498 Rhythm statistics depend on the IET distribution but not on the sequential order of the intervals.
499 Order statistics depend on consecutive intervals and are invariant to uniform rescaling of time.

Table 1: Evaluation practices for graph simulators and LM social simulators. Metric classes are S for snapshot structural, T for text, MT for marginal temporal, D for temporal dependence, L for predictive or likelihood based, and CL for closed loop rollout. Dep. records whether temporal dependence of the generated stream is tested. CL records whether a closed loop test is performed. Every row is explicitly cited.

Reference	System / topic	Class	Dep.	CL
<i>Static graph generation</i>				
You et al. [2018]	GraphRNN, static autoregressive graph generator	S/L	no	no
Bojchevski et al. [2018]	NetGAN, random walk static graph generator	S/L	no	no
Liao et al. [2019]	GRAN, static graph generator	S	no	no
Chanpuriya et al. [2021]	Theory of edge independent graph models	S	no	no
Thompson et al. [2022]	Critique of graph generative evaluation	S	no	no
O’Bray et al. [2022]	Critique of graph generative evaluation	S	no	no
Shirzad et al. [2022]	Graph generative evaluation	S	no	no
<i>Temporal graph generation and benchmarks</i>				
Holme and Saramäki [2012]	Review of temporal networks	MT/D	yes	no
Sizemore and Bassett [2018]	Dynamic graph metrics tutorial	D	partial	no
Zhou et al. [2020]	TagGen, temporal interaction network generator	S	no	no
Zeno et al. [2021]	DYMOND, motif based dynamic network generator	S/MT	partial	no
Zhang et al. [2021]	TG GAN, continuous time temporal graph GAN	S/MT	no	no
Poursafaei et al. [2022]	Evaluation critique for dynamic link prediction	L	partial	no
Gupta and Bedathur [2022]	Survey of temporal graph representation learning and generation	S/L	no	no
Clarkson et al. [2022]	DAMNETS, Markovian network time series generator	S	no	no
Gauvin et al. [2022]	Randomized reference models for temporal networks	D	yes	no
Gupta et al. [2022]	TIGGER	S/MT	no	no
Huang et al. [2023]	TGB benchmark, temporal graph learning	L	no	no
Zhang et al. [2024]	DTGB benchmark, DyTAG discriminative tasks	T/L	no	no
Xiang et al. [2025]	Learning based temporal graph simulator	S	no	no
Du et al. [2025]	GraphMaster, LM agent graph synthesis	S/T	no	no
Chang et al. [2025]	LM generated social networks	S	no	no
Hosseini et al. [2025]	Quality measures for dynamic graph generative models	S	partial	no
<i>LM generated graphs</i>				
Ji et al. [2025]	GAG, LM multi agent DyTAG generator	S/T/MT	no	no
Peng et al. [2025]	GDGB benchmark	S/T/MT	no	no
Ji et al. [2026]	GRAPHIA	S/T	no	no
<i>LM social simulation</i>				
Park et al. [2022]	Social Simulacra, LM populated prototypes	–	no	no
Park et al. [2023]	Generative Agents, Smallville	–	no	no
Gao et al. [2023]	S3, LM social network simulation	T/MT	no	no
Törnberg et al. [2023]	LM social media simulation for feed algorithms	–	no	no
Gao et al. [2024]	Survey of LM empowered agent based modeling	–	no	no
Rossetti et al. [2024]	Y Social, LM social media digital twin	S/T	no	no
Mou et al. [2024]	HiSim, social movement simulation	T/MT	no	no
Yang et al. [2024]	OASIS, million agent social media simulator	T/MT	no	no
Chuang et al. [2024]	LM opinion dynamics	MT	no	no
Piao et al. [2025]	AgentSociety, large scale LM agent society	–	no	no
Wang et al. [2025]	RecAgent, LM user behavior simulator	–	no	no
Laroui and Törnberg [2025]	Critical review of generative social simulations	–	no	no
Liu et al. [2025]	MOSAIC, content dissemination simulation	S/MT	no	no
Zhang et al. [2025]	SocioVerse, world model with a large user pool	–	no	no
Tomašević et al. [2026]	Operational validation on Y Social and Voat	S/T/MT	no	no
Koley [2025]	SALM	S/T	no	no
Zhang et al. [2026b]	POSIM, Hawkes driven public opinion simulator	T/MT	partial	no
Mou et al. [2026]	Survey of LM agent social simulation	–	no	no
Schwager et al. [2026]	Conditioned Comment Prediction, operational validity	T	no	no
Deng et al. [2026]	HawkesLLM, Hawkes and LM cascade simulator	T	no	partial
Miao et al. [2026]	LM multi agent email networks with Hawkes guidance	MT/D/CL	partial	yes
Zhang et al. [2026a]	Hawkes gated crisis response simulator	MT/D	yes	no

500 C.2 Detailed non identification construction and proof

501 The compact statement in Proposition 1 combines the two order sensitivity results used in the main
502 paper. The first result uses the following explicit construction. Let F be any IET distribution with
503 finite variance. Let P_0 generate each actor’s IETs independently from F , and let P_ρ generate them
504 from a stationary Gaussian copula AR(1) process with the same marginal F and parameter $\rho \in (0, 1)$.
505 If partner and text choices are identical and independent of event timing, then P_0 and P_ρ have
506 identical per actor IET marginals, identical burstiness, and identical distributions for every state
507 statistic, while $M(P_\rho) > 0 = M(P_0)$.

508 Let the latent process satisfy

$$z_j = \rho z_{j-1} + \sqrt{1 - \rho^2} \epsilon_j,$$

509 with standard Gaussian innovations, and define

$$\tau_j = F^{-1}(\Phi(z_j)).$$

510 For every value of ρ , the marginal distribution of τ_j is F . The per-actor IET multisets, burstiness, and
 511 every statistic that depends only on the IET marginals therefore have the same distribution under P_0
 512 and P_ρ . Partner and text choices are identical and independent of timing, so the time aggregated graph
 513 and every state statistic also have the same distribution. The map $F^{-1} \circ \Phi$ is non-decreasing. Positive
 514 dependence in the latent AR(1) sequence therefore induces positive dependence between consecutive
 515 IETs for $\rho > 0$, while the dependence vanishes when $\rho = 0$. Hence $M(P_\rho) > 0 = M(P_0)$.

516 C.3 Shuffle invariance

517 **Proof of Proposition 2** The shuffle operator changes the order of an actor's IETs but not the edge
 518 multiset or the IET multiset. It therefore leaves the time aggregated graph and every state or marginal
 519 timing statistic unchanged.

520 For an exchangeable sequence of k IETs with sample mean $\bar{\tau}$ and population sample variance s^2 ,

$$\sum_{i \neq j} (\tau_i - \bar{\tau})(\tau_j - \bar{\tau}) = -ks^2.$$

521 A uniformly selected ordered pair of distinct positions therefore has expected cross product $-s^2/(k -$
 522 $1)$. The expected lag 1 Pearson correlation after a uniform permutation is consequently

$$-\frac{1}{k-1} + o(1/k).$$

523 Pooling adjacent pairs within actors preserves the $O(1/k_u)$ behavior stated in Proposition 2.

524 C.4 Separation under rate changes

525 **Proof of Proposition 3** Under the transformation $\tau \mapsto a\tau$ with $a > 0$, both the mean and standard
 526 deviation of the IETs are multiplied by a . Their ratio in the burstiness coefficient is therefore
 527 unchanged. Pearson correlation is invariant to positive affine rescaling, so memory and lag 1
 528 autocorrelation are also unchanged.

529 The KS distance compares $F_{a\tau}(x) = F_\tau(x/a)$ with the reference distribution and is generally
 530 changed by a . The same argument applies to actor specific KS distances. For the log transformed
 531 Wasserstein distance,

$$\log(a\tau) = \log \tau + \log a.$$

532 Translation of one distribution by c changes the Wasserstein distance by at most $|c|$. Applying this
 533 with $c = \log a$ gives

$$|W_1(\log a\tau, \log \tau') - W_1(\log \tau, \log \tau')| \leq |\log a|.$$

534 Equality holds when the two unscaled log-IET distributions have the same shape.

535 C.5 Finite sample calibration

536 **Proof of Proposition 4** Under exchangeability, the rank of D^{SR} among the $N + 1$ values consisting
 537 of D^{SR} and $D_{1:N}^{RR}$ is uniform. Ties are resolved against the simulator through the greater-than-or-
 538 equal comparison in the definition of q_{RR} . The rank from the upper tail divided by $N + 1$ is therefore
 539 q_{RR} , which gives

$$\Pr(q_{RR} \leq \alpha) = \frac{\lfloor \alpha(N + 1) \rfloor}{N + 1} \leq \alpha.$$

540 The smallest attainable value is $1/(N + 1)$, which occurs exactly when the simulator to real discrep-
 541 ancy exceeds every real to real discrepancy.

Table 2: Synthetic validation of the R2O requirements. The reference stream has memory 0.164 and burstiness 0.464.

Transformation	IET KS	$W_1(\log \tau)$	$ \Delta B $	M
Faithful replicate	0.004	0.009	0.042	0.169
Actor-level shuffle	0.000	0.000	0.000	-0.007
Independent renewal with same marginal	0.006	0.015	0.039	-0.009
Rate multiplied by 20	0.686	2.996	0.000	0.164
Rate multiplied by 20 with shuffle	0.686	2.996	0.000	-0.022

542 C.6 Teacher-forced evaluation

543 **Remark 1.** When observed history is supplied during generation, state and marginal timing errors
544 cannot accumulate in the same way as in a closed loop rollout. Such evaluation can therefore
545 understate errors that arise through feedback over longer horizons. The Order statistics remain
546 informative because the ordering of intervals generated within an evaluated continuation is still
547 produced by the model. R2O itself is independent of the rollout procedure and can be applied to
548 either setting.

549 C.7 Synthetic validation

550 We construct a synthetic stream with 300 actors, lognormal IET marginals, and Gaussian-copula
551 AR(1) dependence chosen to produce memory $M = 0.164$ and burstiness $B = 0.464$. We then apply
552 transformations that isolate the properties used by R2O. The shuffle preserves every marginal timing
553 statistic while removing memory. An independent renewal process with the same IET marginal
554 behaves similarly. A uniform rate change moves the Rate-sensitive distributional statistics while
555 leaving burstiness and memory unchanged.

556 C.8 Control identifiability

557 **Proposition 5 (Control identifiability).** Let Q denote a deterministic or randomized measurement
558 transformation applied by the generation pipeline, such as timestamp quantization or duplicate
559 handling. If a generated process G_θ has the same distribution as the real process E_{real} , then $Q(G_\theta)$
560 and $Q(E_{\text{real}})$ also have the same distribution. A discrepancy that remains after applying the same
561 transformation to both streams cannot be attributed to Q alone. The same logic applies to count
562 conditioning when generated and real actor event counts are matched before scoring.

563 **Proof.** Equality in distribution is preserved under any measurable transformation. Applying the
564 same transformation to both processes therefore preserves equality. The stated interpretation follows
565 from the contrapositive.

566 D R2O calibration and robustness checks

567 D.1 Matched real-window reference

568 For each dataset, the real reference distribution is constructed from pairs of 10,000-edge windows.
569 Candidate pairs are matched on window duration, actor count, edges per actor, and median positive
570 IET. We use disjoint windows where possible and cap reuse at three pairs for any window. Up to 60
571 matched pairs are selected per dataset. When the available stream provides fewer disjoint matches,
572 the empirical reference has correspondingly coarser resolution.

573 For each R2O discrepancy D , the main analysis records both

$$Z_D = \frac{D^{SR} - \mu_D^{RR}}{\sigma_D^{RR}}$$

574 and the empirical tail position q_{RR} defined in the main paper. The standardized score displays
575 the size and direction of a discrepancy in units of real variation. The empirical tail position is

the primary rank-based description. Because matched pairs can share windows, q_{RR} is interpreted descriptively in the empirical analysis. Proposition 4 gives its idealized finite sample interpretation under exchangeability.

Table 3: Matched real-window reference distributions. Windows contain 10,000 edges. Pairs are matched on the covariates described above, with window reuse capped at three pairs. Values report mean and standard deviation of each real to real discrepancy.

Dataset	Windows	Pairs	KS ($\mu \pm \sigma$)	$ \Delta M $ ($\mu \pm \sigma$)	Rhythm ($\mu \pm \sigma$)
dianping	199	60	0.043 ± 0.024	0.027 ± 0.026	0.41 ± 0.11
wikirevision	277	60	0.036 ± 0.019	0.023 ± 0.015	0.36 ± 0.03
weibo_daily	35	52	0.073 ± 0.038	0.035 ± 0.029	0.29 ± 0.02
weibo_tech	10	13	0.075 ± 0.036	0.058 ± 0.046	0.32 ± 0.03

D.2 Timestamp ties

Round-based generation assigns the same timestamp to multiple events and can therefore produce zero IETs. We retain the original stream and evaluate two additional representations. Moment statistics such as memory use a deduplicated representation in which repeated actor-and-timestamp events are removed. Distributional statistics use a jittered representation in which ties are spread within the corresponding generation interval while preserving the order of distinct timestamps.

Across the primary Qwen3-30B-A3B configurations, removing duplicate timestamps explains a median 41% of the absolute burstiness discrepancy. A residual with the same sign remains in all 12 configurations. On two dianping configurations with small original burstiness discrepancies, deduplication increases the discrepancy, indicating that the effect of ties is not uniformly directional.

D.3 Scheduler control

The scheduler control applies the same round quantization used by the generation pipeline to the real stream before computing Order. This produces a real reference subject to the same timestamp resolution as the generated stream. Quantization changes real memory from a 39% increase to a 49% reduction, but the resulting real memory remains above the generated Qwen3-30B-A3B range on all four datasets.

Table 4: Scheduler control for the memory coefficient. Real streams are passed through the same round quantization used by the generation pipeline before memory is recomputed.

Dataset	M^{real}	$M_{\text{quant}}^{\text{real}}$	Reduction	Generated M (min..max)
dianping	0.102	0.052	+49%	-0.276 .. -0.015
wikirevision	0.184	0.144	+22%	-0.011 .. +0.030
weibo_daily	0.159	0.221	-39%	-0.032 .. -0.027
weibo_tech	0.238	0.244	-3%	-0.014 .. -0.009

D.4 Actor-count conditioning

Differences in actor activity may alter marginal IET comparisons even when the timing distribution conditional on activity is similar. We therefore recompute actor level KS after matching generated and real event counts for each actor. The procedure uses 20 subsampling replicates. The median ratio between count conditioned and original KS is 1.006, with values ranging from 0.886 to 1.039 across the primary configurations. The marginal discrepancy therefore remains after event counts are matched.

D.5 Evaluation-window sensitivity

The inherited evaluation uses the first 10,000 edges of each dataset. For wikirevision, this window is unusually early relative to the other candidate windows. We therefore repeat the GPT-4o-Mini L0

and L1 comparison on a representative window selected using the same matching covariates as the real reference construction. Marginal KS decreases substantially and degree MMD also improves, but the Order discrepancy remains. Real memory on the representative window is 0.15, while the generated values are 0.02 and -0.06 for L0 and L1 respectively.

Table 5: Evaluation-window sensitivity on `wikirevision` using GPT-4o-Mini. The representative and inherited initial windows use the same generation and scoring procedures.

Metric	Representative window		Cold-start window	
	L0	L1	L0	L1
Inter event KS (jitter)	0.090	0.088	0.310	0.383
Degree MMD (final)	0.078	0.061	0.151	0.121
M (dedup)	+0.02	-0.06	+0.43	-0.03
Zero IET fraction	0.37	0.30	0.64	0.51

On this representative window, the marginal jitter-arm KS is 0.090 versus 0.310 on the inherited cold-start window, a roughly $3.4\times$ window-specific component of marginal error. Structure is also better, while the dependence deficit persists. Thus the inherited first-10k-edge convention simultaneously understates structural fidelity and overstates marginal timing error.

A particularly transparent Rate example appears on this same representative window: the tool loop worsens the log-duration ratio from 3.83 to 4.44 (approximately $46\times$ to $85\times$ in duration) while structure improves. This illustrates directly that Rate and structural fidelity can move in opposite directions.

D.6 Summary of robustness checks

The controls address distinct alternative explanations. Timestamp handling changes the magnitude of some Rhythm statistics but does not restore Order. Applying the generation scheduler to real data does not account for the missing memory. Matching actor event counts leaves the marginal discrepancy nearly unchanged. Selecting a more representative window reduces marginal error but leaves the dependence discrepancy. These results support the main conclusion that the R2O findings are not explained by the measurement artifacts tested here.

E Complete experimental results

This appendix reports the complete configuration level results underlying the main experiments. Table 6 contains the primary and exploratory configurations. Table 7 gives the matched horizon analysis for the exploratory Qwen3-8B runs. Table 8 reports the timing baselines, and Table 9 reports configuration level quantities used in the robustness analysis.

F Cross system scope and GRAPHIA

GRAPHIA generates activity in fixed one-day bins. An activity predictor determines counts, while the language-model component determines destinations and text; no continuous per event timestamp exists in its generation path. Continuous-time process fidelity is therefore structurally outside its output space. Strong performance under current DyTAG evaluation consequently does not require a continuous time process claim. For the same reason, GRAPHIA cannot provide an independent-codebase replication of the residual event level timing deficit studied here.

G Preregistered confirmatory tests: roll forward and timing locus

The experiments reported in this paper establish that the evaluated simulators do not reproduce the real temporal process and that this failure is separable from structural fidelity. Two confirmatory tests follow from that finding. The *future blind roll forward test* asks whether the temporal deficit grows

Table 6: Complete results for the 16 preregistered configurations and eight exploratory Qwen3-8B configurations. Degree MMD reports structural error; KS is pooled actor level inter-event KS against matched reference slots; Z_{KS} is the same discrepancy in real reference units; M is the memory coefficient from the deduplicated representation, with the real value in parentheses; q_{RR} is the empirical real reference tail position. Superscript s marks a short run; c marks a structural evaluation affected by contamination (weibo_daily L2 excluded; weibo_tech L2 scored on the archived 8.65k-edge run); t marks the 200 event truncated archive, for which the structural evaluation is unavailable and timing uses matched slots at the available horizon; e marks exploratory Qwen3 8B configurations added after registration (L1 runs stopped at round boundaries, 4,900 to 5,696 events, so 8B L0 and L1 rows are scored at different event counts; see Table 7).

Dataset	Backbone	Cell	Deg. MMD	KS	Z_{KS}	M (M^{real})	q_{RR}
dianping	Qwen3-30B-A3B	L0	0.052	0.372	+14.0	-0.01 (0.10)	0.016
		L1	0.026	0.440	+16.9	-0.26 (0.10)	0.016
		L2	0.034	0.433	+16.6	-0.28 (0.10)	0.016
	GPT-4o-Mini Qwen3-8B	L0	0.056	0.340	+12.6	-0.10 (0.10)	0.016
		L0 ^c	0.059	0.444	+17.0	-0.10 (0.10)	0.016
		L1 ^{e,s}	0.045	0.578	+22.7	-0.06 (0.11)	0.016
wikirevision	Qwen3-30B-A3B	L0	0.159	0.456	+22.5	+0.03 (0.20)	0.016
		L1	0.139	0.283	+13.2	+0.00 (0.20)	0.016
		L2	0.115	0.271	+12.6	-0.01 (0.20)	0.016
	GPT-4o-Mini	L0	0.151	0.328	+15.7	+0.43 (0.20)	0.016
		L1	0.121	0.228	+10.3	-0.03 (0.22)	0.016
	Qwen3-8B	L0 ^c	0.131	0.580	+29.1	+0.16 (0.20)	0.016
weibo_daily	Qwen3-30B-A3B	L1 ^{e,s}	0.077	0.257	+11.8	+0.01 (0.25)	0.016
		L0	0.273	0.305	+6.0	-0.03 (0.16)	0.019
		L1	0.112	0.339	+6.9	-0.03 (0.16)	0.019
	Qwen3-8B	L2 ^s	- ^c 0.300	+5.9	-0.03 (0.17)	0.019	0.019
		L0 ^c	0.239	0.299	+5.9	+0.00 (0.16)	0.019
		L1 ^{e,s}	0.137	0.296	+5.8	-0.04 (0.13)	0.019
weibo_tech	Qwen3-30B-A3B	L0	0.209	0.297	+6.2	-0.01 (0.22)	0.071
		L1 ^s	0.097	0.264	+5.3	-0.01 (0.27)	0.071
		L2 ^s	0.214 ^c	0.265	+5.3	-0.01 (0.27)	0.071
	GPT-4o-Mini	L0 ^t	- ^t	0.374	+8.4	-0.04 (0.16)	0.071
	Qwen3-8B	L0 ^c	0.174	0.273	+5.6	+0.01 (0.22)	0.071
		L1 ^{e,s}	0.116	0.331	+7.2	-0.01 (0.23)	0.071

Table 7: Matched-horizon analysis for the exploratory Qwen3-8B configurations. Each L0 stream is truncated to the event count of its paired L1 stream and rescored using the same procedure. KS uses the original generated representation and M uses the deduplicated representation.

Dataset	Arm	n	KS	Z_{KS}	M_{gen}	M_{ref}	$Z_{ \Delta M }$
wikirevision	L0-full	9000	0.580	+29.1	+0.162	0.203	+1.2
wikirevision	L0-trunc	5696	0.416	+20.3	-0.006	0.246	+15.1
wikirevision	L1	5696	0.257	+11.8	+0.008	0.246	+14.2
dianping	L0-full	9000	0.444	+17.0	-0.105	0.102	+6.9
dianping	L0-trunc	5650	0.494	+19.2	-0.079	0.108	+6.2
dianping	L1	5650	0.578	+22.7	-0.060	0.108	+5.4
weibo_daily	L0-full	9000	0.299	+5.9	+0.002	0.157	+4.2
weibo_daily	L0-trunc	5350	0.348	+7.2	-0.016	0.126	+3.7
weibo_daily	L1	5350	0.296	+5.8	-0.043	0.126	+4.7
weibo_tech	L0-full	9000	0.273	+5.6	+0.006	0.225	+3.5
weibo_tech	L0-trunc	4900	0.275	+5.6	-0.012	0.233	+4.1
weibo_tech	L1	4900	0.331	+7.2	-0.011	0.233	+4.1

640 when the simulator is rolled forward on its own generated history instead of being re-anchored to real
641 history at every round. The *timing locus test* asks which component of the agent stack is responsible
642 for the deficit, by comparing where in the stack event timing is decided. Both protocols are frozen
643 and git timestamped. They are protocol only; no results from either test are reported in this paper.

644 G.1 Future blind roll forward test

645 The roll forward test compares paired arms from shared warm-starts. The future blind closed loop
646 arm generates

$$\hat{G}_{t+1} \sim P_{\theta}(\cdot \mid G_{\leq t_c}, \hat{G}_{t_c+1:t}),$$

Table 8: Timing baseline results on the inherited evaluation window. Baselines condition only on the 1,000-edge seed prefix. Baseline values report mean and standard deviation over 20 simulation seeds. Agent configurations are single archived runs scored using the same global-stream timing statistics.

Dataset	Generator	Inter event KS	Burst gap
dianping	seed-IET bootstrap	0.072 ± 0.004	$+0.025 \pm 0.025$
	Hawkes (seed MLE)	0.195 ± 0.030	-0.004 ± 0.018
	Poisson (seed rate)	0.281 ± 0.005	-0.510 ± 0.007
	agent L0	0.823	+0.465
	agent L1	0.559	+0.453
	agent L2	0.580	+0.463
wikirevision	seed-IET bootstrap	0.394 ± 0.004	$+0.302 \pm 0.012$
	Hawkes (seed MLE)	0.342 ± 0.039	$+0.327 \pm 0.097$
	Poisson (seed rate)	0.663 ± 0.004	-0.385 ± 0.007
	agent L0	0.632	+0.563
	agent L1	0.346	+0.590
	agent L2	0.336	+0.590
weibo_daily	seed-IET bootstrap	0.065 ± 0.004	-0.377 ± 0.017
	Hawkes (seed MLE)	0.112 ± 0.000	-0.417 ± 0.015
	Poisson (seed rate)	0.186 ± 0.005	-0.684 ± 0.007
	agent L0	0.602	+0.275
	agent L1	0.532	+0.283
	agent L2	0.538	+0.289
weibo_tech	seed-IET bootstrap	0.122 ± 0.005	-0.315 ± 0.005
	Hawkes (seed MLE)	0.172 ± 0.001	-0.304 ± 0.009
	Poisson (seed rate)	0.306 ± 0.005	-0.540 ± 0.007
	agent L0	0.527	+0.412
	agent L1	0.468	+0.438
	agent L2	0.486	+0.447

where the real continuation after the cutoff is used only for scoring. The closed arm’s actor schedule is recency-weighted sampling from its own generated history and is count-matched by construction.

The teacher forced open-loop arm generates

$$\hat{G}_{t+1} \sim P_{\theta}(\cdot \mid G_{\leq t}),$$

with real post-cutoff history re-injected each step. It is deliberately not future blind.

Define

$$\Delta(h) = S_{\text{temporal}}^{\text{closed}}(h) - S_{\text{temporal}}^{\text{open}}(h).$$

The primary estimand is

$$\text{AUC}_{\Delta},$$

the area under the closed-minus-open temporal-composite gap as a function of horizon. The uncertainty interval is a paired seed level 95% bootstrap interval. A confidence interval containing zero is the preregistered falsifier for the feedback explanation: it indicates temporal-process failure rather than a failure caused by roll forward feedback.

Horizon checkpoints are equally spaced in event count, the protocol’s native clock. A growth-shape parameter is secondary. Seed noise was measured from 24 replicate sets. Paired designs with shared warm-starts are mandatory and require approximately 4 to $5 \times$ fewer runs at $\rho = 0.8$ than an unpaired design. Full-protocol variance is measured using at least three seeds and at least two arms before any full tier sample size is fixed.

G.2 Preregistered temporal composites

Four composites are preregistered:

$$S_{\text{micro}}, \quad S_{\text{temporal}}, \quad S_{\text{macro}}, \quad S_{\text{individual}}.$$

Each is a mean of direction normalized Z scores against the matched real to real band, with larger values indicating worse mismatch.

Table 9: Configuration-level quantities used in the robustness analysis. n is the number of generated events scored. Zero IET is the fraction of actor level inter-event times equal to zero in the generated stream. $\log(T_g/T_r)$ is the log-duration ratio. CC ratio is count conditioned KS divided by original KS. Rhythm is the median actor level KS against the matched real continuation.

Dataset	Backbone	Cell	n	Zero IET	$\log(T_g/T_r)$	CC ratio	Rhythm
dianping	Qwen3-30B-A3B	L0	9000	0.21	+5.11	0.886	0.46
		L1	9000	0.16	+5.21	0.913	0.47
		L2	9000	0.16	+5.22	0.913	0.50
	GPT-4o-Mini Qwen3-8B	L0	9000	0.34	-0.11	1.039	0.44
		L0	9000	0.31	+5.23	0.937	0.56
		L1	5650	0.21	+5.67	0.968	0.61
wikirevision	Qwen3-30B-A3B	L0	9000	0.67	+3.48	1.003	0.67
		L1	9000	0.44	+3.44	1.027	0.42
		L2	9000	0.43	+3.42	1.025	0.43
	GPT-4o-Mini Qwen3-8B	L0	9000	0.64	+3.98	1.003	0.42
		L1	7232	0.51	+3.62	1.003	0.40
		L0	9000	0.87	+4.89	1.004	0.67
weibo_daily	Qwen3-30B-A3B	L1	5696	0.74	+4.15	1.007	0.35
		L0	9000	0.36	+6.50	0.998	0.33
		L1	9000	0.39	+6.34	1.029	0.43
	Qwen3-8B	L2	7500	0.35	+6.72	1.010	0.38
		L0	9000	0.29	+6.52	0.956	0.52
		L1	5350	0.34	+7.08	1.006	0.54
weibo_tech	Qwen3-30B-A3B	L0	9000	0.45	+6.44	1.000	0.33
		L1	7179	0.42	+6.22	1.021	0.34
		L2	7650	0.42	+6.17	1.015	0.33
	GPT-4o-Mini Qwen3-8B	L0	200	0.46	+9.67	1.016	0.62
		L0	9000	0.42	+6.26	1.051	0.50
		L1	4900	0.49	+6.81	1.124	0.43

The temporal composite contains the marginal members $W_1(\log \text{IET})$ and the absolute burstiness gap, together with a separate preregistered temporal subblock consisting of ΔM and node-rhythm divergence. The preregistration originally called the latter pair the “dependence” block. Under the definitions used in this paper, however, only ΔM is order-sensitive; node-rhythm divergence is a marginal timing statistic. We therefore retain the preregistered membership while referring to the pair as the preregistered temporal subblock. Composites are never reported without their component panel.

The archived exploratory results use KS as the headline marginal statistic because the archived scores predate the $\log-W_1$ convention. KS remains a preregistered diagnostic, and confirmatory runs compute both KS and $\log-W_1$.

The confirmatory real to real band uses a block bootstrap preserving actor identity, with at least 30 covariate-matched pairs where disjoint windows suffice. `weibo_tech` has only 13 disjoint pairs. Bootstrap replicates estimate band uncertainty only; they do not add pseudo-observations to the q_{RR} denominator.

G.3 Timing locus test

The timing locus test compares a predeclared menu of timing loci:

1. intensity determined from history;
2. intensity determined from state and history;
3. direct LM emission of Δt ; and
4. a small and large handoff in which the expensive component retains semantic decisions about whom and what, while a lightweight temporal model determines when.

The small and large factorization is not itself claimed as novel. “Handoff wins” is preregistered as an acceptable outcome, including the case in which direct LM emission of Δt loses. Non-inferiority

689 guardrails apply to S_{macro} and edge text similarity. The baseline set also includes a trained neural
690 temporal generator, actor level variants, and cost normalization.

691 **G.4 E-REP stage 2**

692 The confirmatory replication of the window sensitivity result with the local backbone is judged on
693 a preregistered falsifier. The falsifier fires only if both the structural/temporal decoupling and the
694 residual timing gap vanish on the representative window.

695 **H Deviation log**

696 The preregistration carries a git timestamped deviation log. Every entry was committed before the
697 data it governs existed.

698 Entries 1 to 5 concern wording, band construction (including the requirement for a block bootstrap
699 where disjoint real-window pairs are insufficient), Z -scoring of W_1 on log IETs, the two-stage E-REP
700 split, and an E-REP second dataset rule.

701 Entries 6 to 8 fix the roll forward test’s primary estimand (AUC_Δ with uniform weights, directional
702 H1, and paired seed level bootstrap) and make the count versus timing decomposition primary. These
703 entries postdate the exploratory results reported in this paper but precede any roll forward data.

704 Entry 7 makes q_{RR} the primary real reference report, with Z retained for visualization. Applied
705 retroactively to the exploratory results as a robustness view, this choice cannot change any conclusion
706 because all cells are outside the real reference band under both conventions.

707 Entry 9 fixes the closed arm actor schedule as recency-weighted sampling from its own generated
708 history, count-matched by construction.

709 Entry 10 was added after the main results and records the eight exploratory Qwen3-8B cells (L0/L1
710 on all four datasets). They were generated under the same seed prefix, evaluation window, decoding
711 temperature, and scoring pipeline. They are API served with provider reasoning disabled. Retrieval
712 embeddings are computed on CPU and are numerically equivalent. All four L0 lanes completed
713 the full horizon; each L1 lane was stopped at a round boundary (4,900 to 5,696 events) when round
714 latency made the full horizon uneconomical. They are labeled exploratory everywhere and are
715 excluded from every “16-cell” and “12-cell” statement.

716 The full log, including dates, rationales, and affected claims, is included in the supplementary
717 material.

718 **I Extended limitations and scope**

719 All reported results are scored under the lineage’s teacher forced protocol. Because real history is
720 re-injected during generation, state and per-actor marginal errors cannot compound in the same way
721 as under a future blind closed loop rollout. The reported results should therefore be read as optimistic
722 with respect to process error. The closed loop roll forward test is preregistered (Appendix G) but has
723 not yet been run.

724 The empirical claims are scoped to one lineage, one generation harness, four datasets, and 16
725 preregistered cells. The backbone comparison covers the three tested families. The Qwen3-8B arm is
726 exploratory, single-run, and its L0/L1 contrasts are also rescored at matched horizon.

727 Agent configurations are single archived runs rather than replicated generation seeds. Four cells are
728 shorter than the nominal horizon. API served backbones carry provider side serving opacity. The
729 matched real to real calibration quantifies discrepancy relative to natural real-stream variability but
730 does not estimate generation seed uncertainty. The empirical q_{RR} value is a tail position determined
731 by the available real-window reference set and should not be interpreted as a hypothesis test p -value
732 in the empirical analysis.

733 Cross-system generalization is additionally constrained by output-space differences. In particular,
734 GRAPHIA does not expose continuous per event time, so it cannot serve as an independent event level
735 replication of the continuous time residual studied here. Appendix F gives the full scope argument.
736 The preregistration and its deviations are described in Appendix H.

737 **J Broader impact**

738 This work concerns evaluation methodology. Its intended effect is protective: agent-based graph
739 simulators are beginning to be used for forecasting, what if analysis, and synthetic data generation,
740 while the results here show that commonly used validation can miss properties on which those uses
741 depend. Wider adoption of process level evaluation should reduce overtrust in simulator outputs.

742 A possible negative effect is the converse: better temporal-fidelity evaluation could eventually help
743 build simulators of human activity that are harder to distinguish from real traces. The roll forward
744 protocol itself is a detection style instrument and is dual use in the same limited sense as other fidelity
745 metrics.

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Answer: [Yes]

893 Justification: §4 and Appendix A specify datasets, windows, seed prefix, round size, scaffold
894 levels, backbones, serving stacks, decoding regime, baseline fitting, and the arms used for
895 each statistic.

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899 that is necessary to appreciate the results and make sense of them.
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901 material.

902 7. Experiment statistical significance

903 Question: Does the paper report error bars suitably and correctly defined or other appropriate
904 information about the statistical significance of the experiments?

905 Answer: [Yes]

906 Justification: Every cell is normalized against a matched real-vs-real band (Z-scores and
907 exceedance fractions q_{RR} ; Table 3); baseline rows carry $\text{mean} \pm \text{SD}$ over 20 simulation seeds
908 (Table 8); agent cells are single archived runs, stated explicitly in §4, and no per-cell error
909 bars are claimed for them.

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- 911 • The answer [N/A] means that the paper does not include experiments.
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917 run with given experimental conditions).
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922 of the mean.
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932 Question: For each experiment, does the paper provide sufficient information on the com-
933 puter resources (type of compute workers, memory, time of execution) needed to reproduce
934 the experiments?

935 Answer: [Yes]

936 Justification: Appendix A and A.5: archived generation runs from a prior campaign (local
937 vLLM on a single-GPU-class server for Qwen cells; version-pinned commercial API for
938 GPT-4o-Mini cells); the eight new exploratory Qwen3-8B cells ran on a commercial serving
939 API at \$46.78 total with per-event token/latency accounting reported in §4 and Appendix A;
940 all analysis in this paper is CPU-only.

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